

□ 23 ANOVA

Example 1

Is there a difference in the mean amount spent on the different types of goods in the customer satisfaction survey?

This is, to an extent, similar to asking if there a difference in the mean amount spend by males and females or if there is a difference in the mean age of existing and new customers.

In both of these examples we are examining the mean of a scale variable at two levels of a binary variable (i.e. in two independent samples)

Our current example differs in that there are three samples / levels of a nominal variable.

Could we use multiple t tests? I.e. compare the spend on electronics to that on jewellery, electronics to fragrances and finally, jewellery to fragrances?

The problem with this is that as there is a probability of 5% that any one of these tests gives a spurious result. So, this error is inflated to almost $3 \times 5\% = 15\%$ (14.26% to be precise) if we perform multiple tests.

Instead we use a technique called analysis of variance or ANOVA

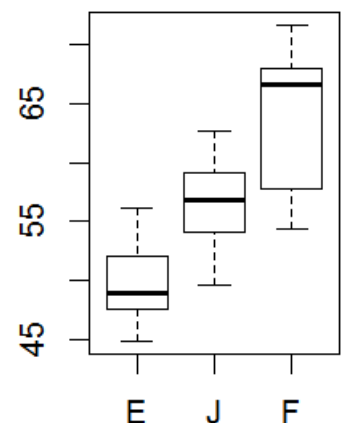
Back to our example

H_0 : The mean spend is the same in all three groups (jewellery, electronics, fragrances)

H_A : The mean spend is **not** the same in all three groups – at least two groups differ

In addition to a clustered boxplot, it is useful to generate some summary statistics (e.g. sample size, mean etc.) for the mean spend in each of the three groups. To this end the function `tapply` is useful.

```
> boxplot(q4_spend ~ q3_goods, names=c("E", "J", "F"))  
  
> tapply(q4_spend, q3_goods, mean, na.rm=T)  
      1      2      3  
49.78250 56.44320 63.82308  
  
> tapply(q4_spend, q3_goods, length)  
      1      2      3  
16 25 14
```



There is certainly a strong indication that the means do differ, and quite possible are different in all three groups.

```
summary(aov(q4_spend ~ as.factor(q3_goods)))  
              Df Sum Sq Mean Sq F value    Pr(>F)      
as.factor(q3_goods)  2 1415.8    707.9    38.6 6.2e-11 ***  
Residuals           51   935.4     18.3                  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
3 observations deleted due to missingness
```

Note that the call to ANOVA (`aov`) is embedded in a call to `summary`. This is how we will always conduct an ANOVA.

The pertinent output is the p value ($\text{Pr}(>F)$), which at 6.2×10^{-11} is < 0.05 and so we have a statistically significant finding – the mean spend is not the same for all groups (i.e. goods types).

Notes

- The ANOVA table does not tell us which groups differ – we must do additional analysis to establish this
- We must indicate in the ANOVA call that `q3_goods` is a nominal, and not a scale variable. We this with the call to `as.factor`.
- It might seem odd that we are using a technique called “analysis of variance” when what we are interested in is an analysis of means.
(The basic idea of ANOVA is to measure the variance in the data in two ways – if the groups had the same means then we would get similar measures of variance. If the two measures differ substantially, then it suggests the groups do not have the same means)
- ANOVA can be carried out with just two samples – but the results will be identical to that found with an independent samples t test. And the latter may be preferable as it reports a confidence interval.

Example 2

An animal feeds firm blends a single item and has two production lines in operation on which there are a multitude of workers. Grain is the primary raw material used by the firm and this is supplied by one of Great Grains, Grain Chain or Kelly’s Food. Part of the quality control process requires that the finished item have its calorie content determined. It is of interest to know if calorie content differs by raw material supplier. Suitable random samples from each of the raw material suppliers are collected and blindly analysed in random order. The data (calories / 100g) is below.

Great Grains	G	124	123	131	146	136	114	132	122	124	138	129	128
Grain Chain	C	146	156	143	145	156	133	143	150	138	152	155	151
Kelly’s Food	K	116	122	112	126	116	121	125	123	121	132	112	111

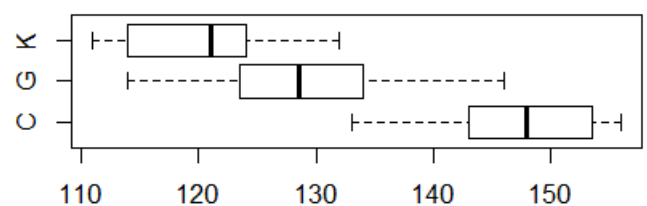
The data needs to be arranged in stacked format and read in to R.
(Below G, C and K rather than the full brand names are used)

Both an examination of the sample means and the boxplot suggest that there is a difference

```
> tapply(cal, brand, mean)
      C      G      K
147.3333 128.9167 119.7500
```

```
boxplot(cal~brand, horizontal = T)
```

brand	cal
Great Grains	124
Grain Chain	146
Kelly’s Food	116
Great Grains	123
Grain Chain	156
:	:



The ANOVA table below yields a p value of < 0.05 – thus indicating a difference in calorie content measure between suppliers.

```
summary(aov(cal~brand))
              Df Sum Sq Mean Sq F value    Pr(>F)    
brand           2   4736   2368.1    42.43 7.55e-10 ***
Residuals      33   1842     55.8                
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note that a call to `as.factor` is not necessary here as the brands are not coded with numbers

Note too that Df (degrees of freedom) for brand is 2 – this should be one less than the number of brands. Also, total Df (2 + 33) should be one less than the total sample size. I.e. 2 + 33 + 1 = 36 (= 3 x 12)

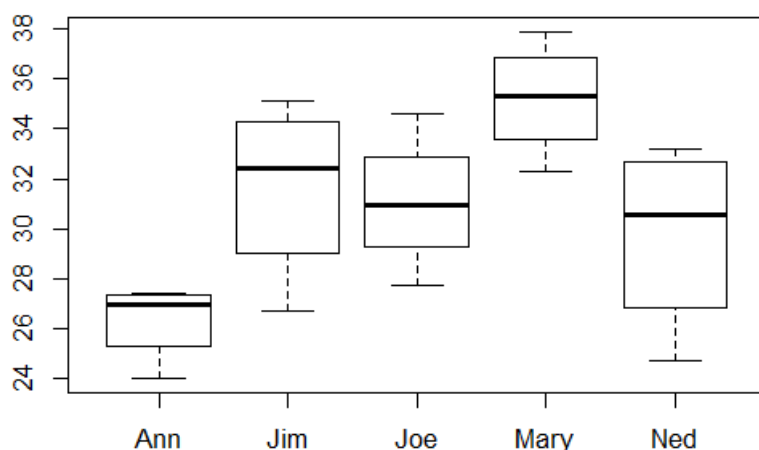
Example 3

It is required to determine if the time taken by the five workers Ann, Joe, Jim, Mary and Ned to assemble one item at a station differs. Each worker assembles four items with the run order randomised. The data is below.

run	1	2	3	4	5	6	7	8	9	10
worker	Ann	Joe	Ann	Joe	Ann	Ann	Mary	Jim	Ned	Mary
time	24.0	31.1	27.4	27.7	26.6	27.3	35.8	35.1	24.7	32.3
run	11	12	13	14	15	16	17	18	19	20
worker	Mary	Joe	Mary	Joe	Ned	Ned	Jim	Ned	Jim	Jim
time	34.8	30.8	37.9	34.6	32.2	28.9	26.7	33.2	31.3	33.5

```
> tapply(time, worker, mean)
  Ann   Jim   Joe   Mary   Ned 
26.325 31.650 31.050 35.200 29.750

> boxplot(time~worker, horizontal = T)
```



```
> summary(aov(time~worker))
              Df Sum Sq Mean Sq F value Pr(>F)
worker         4  165.1   41.27   4.698 0.0117 *
Residuals     15  131.8    8.79
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

So again, we reject the null hypothesis that assembly times are the same for all workers

Post hoc tests

In the last example, we might speculate that Ann has the shortest time, Mary the longest, and the other three are about the same etc.

A better way to proceed is to use a post hoc test. There are a plethora of such tests available – we will consider just one, Tukey's honest significance difference.

We need to decouple the calls to `aov` and to `summary` to do so.

```
> asem.times = aov(time~worker)
> summary(asem.times)
```

```
> TukeyHSD(asem.times)
```

The first two calls generate the ANOVA table as before. The output from the third call is below

```
TukeyHSD(asem.times)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = time ~ worker)

$worker
      diff      lwr      upr      p adj
Jim-Ann  5.325 -1.146824 11.796824 0.1333014
Joe-Ann  4.725 -1.746824 11.196824 0.2132110
Mary-Ann  8.875  2.403176 15.346824 0.0055218
Ned-Ann  3.425 -3.046824  9.896824 0.4997321
Joe-Jim  -0.600 -7.071824  5.871824 0.9983563
Mary-Jim  3.550 -2.921824 10.021824 0.4661596
Ned-Jim  -1.900 -8.371824  4.571824 0.8899883
Mary-Joe  4.150 -2.321824 10.621824 0.3210696
Ned-Joe  -1.300 -7.771824  5.171824 0.9695117
Ned-Mary -5.450 -11.921824  1.021824 0.1203403
```

The p value for a comparison of Mary and Ann is $0.0055218 < 0.05$, so we can say there is a statistically significant difference between this pair and [2.4, 15.3] minutes is a 95% confidence interval for the mean difference in assembly time for the two.

None of the other pair-wise difference are statistically significant.

Exercise

Show that the following post hoc output is obtained for our first problem and interpret what it indicates.

```
Tukey multiple comparisons of means
  95% family-wise confidence level

Fit: aov(formula = q4_spend ~ as.factor(q3_goods))

$`as.factor(q3_goods)`
      diff      lwr      upr      p adj
2-1  6.660700  3.350895  9.970505 3.44e-05
3-1 14.040577 10.180395 17.900759 0.00e+00
3-2  7.379877  3.844868 10.914886 1.84e-05
```

Exercise

Write the R code that generates the output below for the second problem (below) and interpret.

```
Tukey multiple comparisons of means
  95% family-wise confidence level

Fit: aov(formula = cal ~ brand)

$brand
      diff      lwr      upr      p adj
G-C -18.416667 -25.90061 -10.932719 0.0000025
K-C -27.583333 -35.06728 -20.099385 0.0000000
K-G  -9.166667 -16.65061  -1.682719 0.0135905
```